

03_modern_data_tools_introduction

September 5, 2025

1 Lesson 3: Modern Data Tools with AI Integration

1.1 AI-Assisted Performance Optimization and Interactive Dashboards

Weeks 5–6

- **Week 5 (2–3 hours):** AI-guided tool selection and performance optimization for materials science datasets.
- **Week 6 (3–4 hours):** AI-assisted interactive dashboard creation and deployment strategies.

Course Alignment: Builds on AI-assisted data processing skills (Lesson 2.5). This lesson emphasizes **AI-guided tool selection, performance optimization, and interactive dashboard creation** for materials science, with applications to **large datasets, real-time analysis, and research communication**.

1.2 Week 5 — AI-Guided Tool Selection and Performance Optimization (2–3 hrs)

1.2.1 Learning Objectives

- **Use AI to assess data processing needs** and recommend appropriate tools for materials science datasets
- **Apply AI-assisted performance benchmarking** with iterative optimization strategies
- **Design data processing workflows** using AI consultation and validation
- **Reflect on AI limitations** in tool recommendations and performance optimization advice

1.2.2 Prerequisites

Complete Lesson 2.5 - You should have: - Experience with AI-assisted data processing strategies - Understanding of performance metrics and optimization principles - Basic knowledge of when to use different data processing approaches - Working AI tools (Claude, ChatGPT, or GitHub Copilot)

AI Tools Ready: Claude, ChatGPT, or GitHub Copilot configured and accessible

1.2.3 Mini-Lecture (~25–30 min)

Modern data tools offer significant performance advantages over traditional pandas, but choosing the right tool requires understanding your specific materials science requirements. AI can help guide these decisions by analyzing your data characteristics, performance needs, and research goals.

Core Principles for AI-Guided Tool Selection

1. **Data-First Analysis** — Let AI assess your dataset characteristics before tool selection

2. **Performance Requirements** — Use AI to define specific performance targets and constraints
3. **Iterative Optimization** — Test AI recommendations, measure results, refine approach
4. **Materials Science Context** — Ensure AI considers domain-specific requirements and constraints

AI Prompt Patterns for Tool Selection

Pattern 1: Tool Assessment

ROLE: Data engineering consultant

AUDIENCE: Materials science research team

GOAL: Recommend optimal data processing tools for [specific materials dataset]

INPUTS: [Dataset size, data types, analysis requirements, hardware constraints]

CONSTRAINTS: 200 words; include performance estimates; suggest 2-3 tool options

TASK: Analyze requirements and recommend processing approach

EVAL CRITERIA: Specific, performance-focused, considers materials science context

Pattern 2: Performance Optimization

ROLE: Performance optimization specialist

AUDIENCE: Research computing team

GOAL: Optimize data processing performance for [specific analysis task]

INPUTS: [Current approach, performance metrics, bottlenecks, requirements]

CONSTRAINTS: 250 words; suggest specific optimizations; include trade-offs

TASK: Identify bottlenecks and recommend improvements

EVAL CRITERIA: Specific, measurable, considers resource constraints

Pattern 3: Workflow Design

ROLE: Data pipeline architect

AUDIENCE: Materials engineering team

GOAL: Design optimal data processing pipeline for [research objective]

INPUTS: [Data characteristics, analysis goals, performance requirements]

CONSTRAINTS: 300 words; include step-by-step process; note potential bottlenecks

TASK: Create processing pipeline with tool recommendations

EVAL CRITERIA: Feasible, efficient, materials-science appropriate

Modern Data Tools for Materials Science

Polars — Fast, memory-efficient alternative to pandas - **Best for:** Large datasets (>100MB), complex aggregations, memory-constrained systems - **Materials applications:** Property databases, composition analysis, statistical processing - **AI can help with:** Syntax conversion, optimization strategies, performance tuning

Dask — Parallel computing for larger-than-memory datasets - **Best for:** Very large datasets (>1GB), distributed computing, complex workflows - **Materials applications:** High-resolution microscopy, extensive property databases - **AI can help with:** Workflow design, resource allocation, optimization strategies

Streamlit — Interactive dashboard creation - **Best for:** Research communication, interactive data exploration, stakeholder presentations - **Materials applications:** Property visualization, composition optimization, results sharing - **AI can help with:** UI design, functionality planning, deployment strategies

When AI Recommendations May Be Limited - Hardware-specific constraints: AI may not know your exact system capabilities - **Domain-specific requirements:** May miss materials science-specific considerations - **Performance trade-offs:** May not fully understand your specific priorities - **Tool ecosystem:** May not be aware of all available options or recent updates

1.3 Week 5 — Grading Rubric (10 pts)

GuidedActivity	Criterion	Excellent (full)	Partial	Pts
1	AI tool assessment	Clear requirements analysis, specific tool recommendations, materials context	Vague or generic suggestions	4
1	Performance optimization	2+ iterations with KEEP/CHANGE notes, measurable improvements	1 iteration or superficial changes	3
2	Workflow implementation	Functional code, AI-guided optimization, performance validation	Basic implementation without AI integration	3

Submission: This notebook with prompts, outputs, reflections, and analysis in the Guided Activities below.

Reminder: Each reflection cell tells you the word count requirement and what will be graded. Aim to stay within the word limits, but don't worry if you exceed them slightly.

1.3.1 Guided Activity 1: AI-Assisted Tool Selection and Performance Optimization (60–75 min)

Your Tasks

1. **Choose a materials science dataset** (use provided sample or your own)
2. **Use AI to assess tool requirements** and recommend processing approach
3. **Implement AI-suggested optimizations** and measure performance improvements
4. **Iterate on the approach** based on AI feedback and performance results
5. **Reflect on AI recommendations** and validation strategies

Materials Science Dataset Options: - **Large property database:** >10,000 alloy compositions with mechanical properties - **High-resolution microscopy data:** Image analysis results with particle distributions - **Time-series corrosion data:** Environmental exposure measurements over

time - **Thermal analysis data:** DSC/TGA curves with multiple heating rates - **Composition optimization data:** Multi-element alloy design space exploration

Tip: Use the prompt patterns above and make sure to include specific performance requirements and materials science context in your prompts.

```
[ ]: # === Dataset Selection and Setup ===
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import time
import psutil
import textwrap

# Choose your dataset (uncomment one option or use your own)

# Option 1: Large property database (sample)
np.random.seed(42)
n_samples = 15000 # Large dataset for performance testing
data = pd.DataFrame({
    'Alloy_ID': [f'Alloy_{i:05d}' for i in range(1, n_samples + 1)],
    'Al_content': np.random.uniform(85, 99, n_samples),
    'Cu_content': np.random.uniform(0, 5, n_samples),
    'Mg_content': np.random.uniform(0, 3, n_samples),
    'Si_content': np.random.uniform(0, 2, n_samples),
    'Fe_content': np.random.uniform(0, 1, n_samples),
    'Tensile_Strength_MPa': np.random.uniform(200, 600, n_samples),
    'Yield_Strength_MPa': np.random.uniform(150, 500, n_samples),
    'Elongation_Percent': np.random.uniform(2, 25, n_samples),
    'Hardness_HV': np.random.uniform(50, 200, n_samples),
    'Density_g_cm3': np.random.uniform(2.6, 2.9, n_samples),
    'Processing_Temp_C': np.random.uniform(400, 600, n_samples),
    'Aging_Time_hours': np.random.uniform(1, 24, n_samples)
})
dataset_type = "Large aluminum alloy property database"
research_question = "Optimize alloy composition for high strength-to-weight_
ratio while maintaining corrosion resistance"

# Option 2: High-resolution microscopy data (sample)
# n_particles = 50000
# data = pd.DataFrame({
#     'Particle_ID': range(1, n_particles + 1),
#     'Area_pixels': np.random.lognormal(4, 1, n_particles),
#     'Perimeter_pixels': np.random.lognormal(3, 0.8, n_particles),
#     'Eccentricity': np.random.beta(2, 5, n_particles),
#     'X_position': np.random.uniform(0, 2048, n_particles),
#     'Y_position': np.random.uniform(0, 2048, n_particles),
#     'Phase': np.random.choice(['Alpha', 'Beta', 'Gamma'], n_particles)
```

```
# })
# dataset_type = "High-resolution microscopy particle analysis data"
# research_question = "Analyze particle size distribution and phase
↳relationships"

print(f"Dataset: {dataset_type}")
print(f"Research Question: {research_question}")
print(f"Dataset shape: {data.shape}")
print(f"Memory usage: {data.memory_usage(deep=True).sum() / 1024 / 1024:.1f}
↳MB")
print(f"\nFirst few rows:")
print(data.head())
```

```
[ ]: # === Step 1: AI Tool Assessment ===

# Create your tool assessment prompt
prompt_tool_assessment = f"""
ROLE: Data engineering consultant
AUDIENCE: Materials science research team
GOAL: Recommend optimal data processing tools for {dataset_type}
INPUTS: Dataset has {data.shape[0]} rows, {data.shape[1]} columns, {data.
↳memory_usage(deep=True).sum() / 1024 / 1024:.1f} MB memory usage. Research
↳question: {research_question}. Need to analyze composition-property
↳relationships and optimize alloy design.
CONSTRAINTS: 200 words; include performance estimates; suggest 2-3 tool options
TASK: Analyze requirements and recommend processing approach
EVAL CRITERIA: Specific, performance-focused, considers materials science
↳context
"""

print("Tool Assessment Prompt:")
print(textwrap.indent(prompt_tool_assessment, "    "))

# Copy this prompt to your AI tool and paste the response below
response_tool_assessment = """
[Paste AI response here]
"""

print("\nAI Response:")
print(textwrap.indent(response_tool_assessment, "    "))
```

1.3.2 Guided Activity 2: AI-Assisted Performance Implementation (45-60 min)

Your Tasks

1. Implement AI-recommended tools (Polars, Dask, or other suggestions)
2. Use AI to optimize performance with specific improvements
3. Benchmark performance against baseline pandas approach

4. **Iterate on optimizations** based on AI feedback and results
5. **Reflect on AI performance advice** and its effectiveness

Performance Metrics to Track: - **Processing time:** Time for data loading, cleaning, and analysis - **Memory usage:** Peak memory consumption during operations - **Scalability:** Performance with different dataset sizes - **Accuracy:** Correctness of results compared to baseline

Tip: Use the performance optimization prompt pattern and be specific about your hardware constraints and performance targets.

1.4 Week 6 — AI-Assisted Interactive Dashboard Creation (3–4 hrs)

1.4.1 Learning Objectives

- **Use AI to design dashboard functionality** and user interface for materials science data
- **Apply AI-guided Streamlit development** with iterative improvement
- **Create interactive visualizations** using AI consultation and optimization
- **Deploy and test dashboards** with AI-assisted troubleshooting and refinement

1.4.2 Workflow & Timeboxing

Step 1 — Dashboard Design with AI (30–40 min)

- Use AI to analyze requirements and design dashboard functionality
- Create wireframes and user flow with AI assistance
- Define interactive elements and data visualizations

Step 2 — AI-Assisted Development (90–120 min)

- Use AI to generate Streamlit code and optimize implementation
- Iterate on functionality based on AI feedback and testing
- Apply AI-guided performance optimization and error handling

Step 3 — Testing and Refinement (30–40 min)

- Use AI to identify bugs and suggest improvements
- Test dashboard with different datasets and user scenarios
- Apply AI-assisted deployment strategies and troubleshooting

Step 4 — Reflection and Documentation (20–30 min)

- Reflect on AI assistance effectiveness and limitations
- Document lessons learned and best practices
- Plan future dashboard enhancements with AI guidance

1.5 Week 6 — Grading Rubric (15 pts)

Criterion	Excellent (full)	Partial	Pts
Dashboard design	Clear AI-guided requirements analysis, specific functionality, materials context	Vague or generic design	4

Criterion	Excellent (full)	Partial	Pts
AI-assisted development	2+ iterations with AI feedback, functional code, optimization	Basic implementation without AI integration	5
Performance & testing	Comprehensive testing, performance validation, error handling	Limited testing or performance issues	3
Reflection & documentation	Specific AI limitations, actionable insights, clear documentation	Generic statements or poor documentation	3

Submission: This notebook with prompts, outputs, code, and reflections in the Guided Activities below.

Reminder: Each reflection cell tells you the word count requirement and what will be graded. Aim to stay within the word limits, but don't worry if you exceed them slightly.

1.6 Synthesis (5–10 min)

Write 3 bullets you will **carry forward** to future data processing and dashboard projects: - Best AI prompt pattern for tool selection and optimization - Most important performance metric to track for your research - Most effective AI-assisted development workflow for interactive tools

1.7 Next Steps

In the next lesson, we'll learn about **basic statistical analysis with AI assistance**, including hypothesis testing, experimental design, and automated reporting for materials science data.

Remember: Modern data tools are most effective when combined with AI guidance. Use AI as a consultant for tool selection, optimization, and development, but always validate recommendations with your domain knowledge and experimental data.

1.8 Resources and References

1.8.1 Modern Data Tools

- [Polars Documentation](#)
- [Dask Documentation](#)
- [Streamlit Documentation](#)

1.8.2 AI-Assisted Development

- [AI Tools for Data Science](#)
- [Prompt Engineering for Developers](#)

- [AI-Assisted Code Generation](#)

1.8.3 Materials Science Applications

- [Materials Data Science Workflows](#)
- [Interactive Materials Visualization](#)
- [Materials Property Databases](#)

Happy modern data processing!